SonoSPADE: Real-Time, Per-Tissue Ultrasound Texture Synthesis for Closed-Loop Acquisition

Anonymous

Anonymous Institution

Abstract. Closed-loop simulators for learning ultrasound *acquisition* render a new B-mode at every agent step, so their texture stage must be real-time and controllable: realism-first diffusion is orders of magnitude too slow to run in the loop, and global sim-to-real translators impose one appearance map with no per-tissue control. We present *SonoSPADE*, a physics-conditioned, *per-tissue* ultrasound texture stage, real-time and GPU-free, trained unpaired with no real annotations: a generator conditioned on *both* a physics render and the tissue label slice, supervised through a segmenter pretrained for free on aligned simulator pairs and frozen to pseudo-label the real pool for an OASIS-style discriminator and a per-tissue consistency loss. A single pass runs at 15 to 85 frames per second *without a GPU*, two to three orders of magnitude faster than diffusion. Our primary evidence is per-tissue: on the public Kaggle abdominal set, relabeling a tissue changes its own texture far more inside its region than outside (label-swap locality ≈ 26, versus 0 for every label-free translator), per-organ intensity matches real roughly $2\times$ better than any learned baseline, and structure is preserved $2.4\times$ better; on whole-frame FID/KID it trails by design, as those metrics reward the global recolor it does not perform. As a first in-the-loop proof of concept, on an organ-conditioned acquisition task scored by a *fixed* real-trained organ recognizer shared across texture stages, the per-tissue reward yields the highest ground-truth liver visibility in all three seeds — a small, seed-consistent, liver-only margin. Code and the reproducible pipeline are released.

Keywords: Ultrasound synthesis · Semantic image synthesis · Real-time synthesis · Closed-loop simulation · Sim-to-real.

1 Introduction

Learning ultrasound *acquisition* in simulation, where an agent moves a virtual probe and observes the B-mode it would produce, needs a texture stage that renders a new frame at every step: it must be real-time and run on commodity hardware, and controllable so the agent's actions map to anatomy. Realism-first synthesizers do not meet this. Diffusion produces high-fidelity ultrasound but at seconds per frame [7], orders of magnitude too slow to sit in a loop that consumes millions of frames; global sim-to-real translators run on the grayscale image alone [6], imposing one appearance map with no per-tissue control, so they cannot guarantee that liver reads as liver or kidney as kidney. Yet the pipeline already knows the anatomy: a canonical label slice drives a physics B-mode with acoustic shadowing, a depth-varying point spread function, and correlated speckle. We make the learned stage *per tissue* by conditioning on that label slice and keep it single-pass so it runs in real time without a GPU, improving on both semantic image synthesis [4] and segmentation-consistent CT-to-US translation [6]. Realism-first anatomy-consistent synthesizers [11] target offline augmentation at higher fidelity; we trade fidelity for the speed and per-tissue control a texture stage needs to run in the loop.

Two design choices set SonoSPADE apart: the generator is conditioned on *both* the physics render and the label (refining a tissue-correct render rather than inventing ultrasound from CT, the simulator's unique lever), and a frozen segmenter pretrained free on aligned pairs supplies both the OASIS-style discriminator's pseudo-labels [8] and a per-tissue consistency term, removing SPADE's [4] paired-real-data requirement; a one-sided contrastive backbone [3] keeps it laptop-trainable without CUDA. Our contributions are:

- A real-time, GPU-free per-tissue texture stage: a single-pass generator at 15–85 frames/s on a laptop with no discrete GPU, two to three orders of magnitude faster than diffusion, that runs at every step of a closed-loop acquisition simulator without becoming the bottleneck — to our knowledge the first per-tissue texture stage that runs *in the loop* at real-time rates.
- A physics-conditioned dual-input generator conditioned on *both* a physics render and the label slice, giving per-tissue control (relabel a region and only its texture changes) that global translators lack, trained with free unpaired segmentation consistency: a frozen simulator-pretrained segmenter supplies the OASIS discriminator's pseudo-labels and a per-tissue consistency loss, so no real labels are needed.
- A per-tissue numeric evaluation (per-class Wasserstein, a label-swap locality metric, speckle SNR/GLCM) addressing a stated open problem, plus a diagnosis and two drop-in fixes for the near-zero downstream transfer (a linear-vs-curvilinear geometry gap, not texture): a sector remap and test-time adaptation lift TSTR severalfold, moving SonoSPADE from worst to overtaking the raw render on every seed.
- A first in-the-loop proof of concept: on an organ-conditioned task scored by a *fixed* real-trained recognizer shared across texture stages, the per-tissue reward gives the highest ground-truth liver visibility in all three seeds — a small, liver-only margin. The primary contributions are the offline per-tissue results.

2 Related Work

Semantic synthesis and label-consistent US. SPADE conditions generation on a label map via spatially-adaptive normalization [4] and OASIS recasts

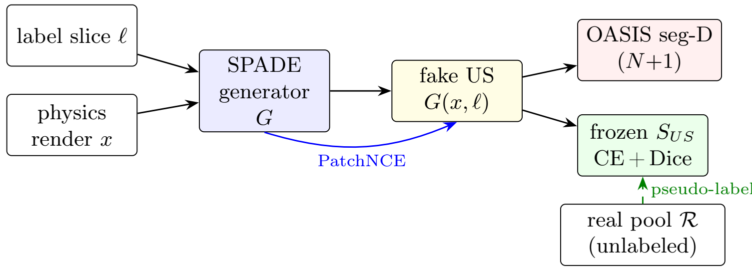

Fig. 1. SonoSPADE. The label slice and the physics render both condition the generator. A segmenter S_{US} pretrained free on aligned simulator pairs is frozen to pseudo-label the real pool for an OASIS (N+1) discriminator and to enforce a per-tissue consistency loss; PatchNCE preserves content. No inverse generator, no cycle loss.

the critic as a per-pixel segmentation discriminator [8]; both are supervised and label-only. We borrow the adaptive conditioning but drive it from a physics render and supervise the discriminator with *free* pseudo-labels. A parallel line generates label-consistent US from anatomical models to train segmenters: e.g. Gilbert et al. translate model renders to echo with a CycleGAN (87–91 Dice on cardiac structures) [1]; these global translators render single structures well but, being whole-image maps, offer no per-tissue control, which we expose by conditioning on the label. **Anatomy-consistent US synthesis.** Closest to us, Vitale et al., on the same abdominal data, combine a segmentation-guided loss and polar (sector) training to suppress hallucinations and sharply improve realism [11]; related structure-preserving translators include S-CycleGAN [6,14] and CACTUSS [9]. Our free segmentation consistency and sector remap build on this recipe; our additions are orthogonal — per-tissue *conditioning* (not merely a loss), a label-swap locality metric, and, centrally, single-pass GPU-free inference for in-the-loop use. These methods optimize offline realism; we optimize per-frame latency so the stage can run inside an acquisition loop. Semantic diffusion [7] reaches higher fidelity but needs tens to hundreds of passes per frame (seconds of compute), prohibitive for an agent that consumes millions of frames.

3 Method

Fig. 1 summarizes SonoSPADE. Let x be a physics B-mode render and ℓ its one-hot canonical tissue label slice (K=13 classes). The generator $G(x, \ell)$ refines x into a realistic frame; it is trained unpaired against a real pool $\mathcal{R}$ with three forces.

3.1 Physics-conditioned dual-input generator

G is a ResNet encoder-decoder whose residual and upsampling blocks are SPADE-normalized: a parameter-free instance norm whose per-pixel scale and bias are predicted from the resized label map ℓ, so the semantic layout modulates activations at every scale [4]. The physics render x is the content stream, so G refines a tissue-correct render rather than hallucinating anatomy. Content is preserved by multilayer PatchNCE on G's own encoder features [3] between x and $G(x, \ell)$; there is no inverse generator and no cycle loss.

3.2 Free unpaired segmentation consistency

The simulator emits perfectly aligned (x, ℓ) pairs, so a compact U-Net [5] segmenter S_{US} is pretrained at no annotation cost, then *frozen* and used two ways. (i) It pseudo-labels the unlabeled real pool, giving an OASIS-style $(N+1)$-class per-pixel discriminator D a semantic target for free [8]: G wins only when each synthesized pixel reads as its conditioned tissue, a per-pixel realism signal. (ii) A consistency loss $\mathcal{L}_{seg} = \text{CE}+\text{Dice}$ requires $S_{US}(G(x, \ell))$ to match ℓ, so G cannot collapse to a global recolor. The full objective is

$$\mathcal{L} = \mathcal{L}_{adv}^{\text{OASIS}} + \lambda_{nce}\mathcal{L}_{nce} + \lambda_{seg}\mathcal{L}_{seg} + \lambda_{tc}\mathcal{L}_{tc}, \tag{1}$$

with class-balanced per-pixel cross entropy in $\mathcal{L}_{adv}$ so rare tissues are not ignored.

3.3 Grouped scheme and training

Fat, GI wall, pancreas, muscle, and generic soft tissue have near-equal backscatter and render near-identically, so S_{US} and D operate in a *grouped* scheme (these five merge; distinct structures keep their class) while G still conditions on the full K classes. Three light stabilizers help: a temporal-consistency term $\mathcal{L}_{tc}$ requiring G to commute with a small in-plane shift, an OASIS-style LabelMix on D [13], and an R1 gradient penalty [2] (average pooling keeps D twice-differentiable); the deployed generator is an EMA of the training weights.

4 Experiments

4.1 Setup

We build on a commodity-laptop ultrasound substrate that renders a physics B-mode from a CT-derived canonical label volume (13 tissues via TotalSegmentator [12]) and exposes a label-conditioned translator interface. Free aligned (x, ℓ) pairs come from eight abdominal CT cases with pose sweeps and domain randomization. For the real head-to-head we use the public Kaggle abdominal ultrasound set [10]: every translator is trained on its 404-frame real *train* split, and evaluated on the disjoint 60 annotated real frames (liver, kidney, gallbladder, spleen, vessels), a leakage-free real test. All translators are evaluated on the same held-out content. Everything runs on Apple Silicon (MPS, no CUDA).

4.2 The label conditioning is genuinely per-tissue

The central claim is that texture is per-tissue, not a global recolor. We measure this directly: holding the physics render fixed, we relabel a tissue region and measure the texture change inside versus outside. A global recolor scores near zero inside; SonoSPADE changes the region's own texture 0.065 inside versus 0.005 outside (ratio ≈ 13), and the complementary label-swap locality metric (Table 1) gives **26.3** versus 0 for every label-free translator by construction (Appendix Fig. 2). Per-class speckle SNR also shifts per tissue, confirming distinct learned statistics, not a uniform map.

4.3 Free segmentation supervision

S_{US} pretrained on the aligned pairs reaches 0.93/0.90/0.72 Dice on liver/lung/vessels (acoustically distinct) on held-out patients. The canonical 13-class macro Dice is capped at 0.50 by degenerate soft tissues; merging them (grouped scheme) lifts it to 0.66 (merged soft-tissue class 0.75 vs 0.35), with no change to distinct tissues — an honest reflection of B-mode acoustics, not a tuning artifact.

4.4 Comparison to baselines

Table 1 compares raw physics, CycleGAN [14], CUT [3], an S-CycleGAN-style variant [6], and SonoSPADE on the real Kaggle set. **Structure:** SonoSPADE preserves anatomy far better than any learned translator (edge IoU 0.41 vs 0.17/0.11 for CUT/CycleGAN, a 2.4× margin) — trained to match real appearance, the tissue-agnostic translators aggressively remap the frame and drift on structure, while physics-plus-label conditioning keeps anatomy intact. **Per-organ realism:** we report intensity W_1 *restricted to the annotated organ masks*, since the whole-frame score is dominated by the background a global recolor matches trivially. Inside the masks the recolors' per-tissue fidelity collapses (organ W_1 0.266/0.246 vs 0.064 for the render they started from), whereas SonoSPADE keeps it (**0.119**, $\approx 2\times$ better than any learned baseline and closest to raw physics), and is the only method with per-tissue control (locality **26.3** vs 0). **TSTR** is near zero for all linear methods (SonoSPADE lowest, 0.025; kidney/spleen Dice 0): the score is dominated by the linear-vs-curvilinear geometry gap no method addresses, and is further depressed for SonoSPADE because its conditioning shifts features from what the pretrained classifier expects. Matching the sector geometry and adding test-time adaptation resolves most of it and reverses the ranking (next).

Whole-frame realism (FID/KID). SonoSPADE does *not* win whole-frame FID/KID (Appendix Table 4): its scores sit near the untranslated render and above the tissue-agnostic translators. This is the organ-W_1 story from the other side — FID/KID score *overall* appearance, which a global tone map matches cheaply and which is blind to whether each organ reads correctly. Matching the sector geometry improves all absolute scores (a large confound) but not the ranking;

Table 1. Baseline comparison on the real Kaggle set. Edge IoU (structure vs the physics input, higher better); *organ* intensity W_1 (mean over the five annotated organs, excluding the background a global recolor trivially matches; lower better); label-swap locality (per-tissue control, higher better; 0 for label-free translators by construction); and TSTR Dice. SonoSPADE preserves structure 2.4× better than the best baseline and per-organ intensity roughly 2× better than any learned translator, and is the only method with per-tissue control. TSTR is near zero for all linear methods, dominated by the sim-to-clinical geometry gap (see text); the SonoSPADE-curvi row renders the same content in the matched sector geometry (Experiment A, Sec. 4.6), and its TSTR is the five-seed mean from Table 2.

Method	Edge IoU ↑	Organ W_1 ↓	Locality ↑	TSTR Dice ↑
Raw physics	**1.00**	0.064	0.0	0.049
CycleGAN	0.11	0.266	0.0	0.032
CUT	0.17	0.246	0.0	0.031
S-CycleGAN (reimpl.)	0.13	0.223	0.0	0.026
SonoSPADE	0.41	**0.119**	**26.3**	0.025
SonoSPADE-curvi	0.54	0.115	18.6	0.099

SonoSPADE's advantage is per-organ fidelity and per-tissue control, and we do not optimize FID.

4.5 Ablation of the training terms

Ablating SonoSPADE's terms one at a time (retrained at a fixed shared budget; Appendix Table 5) shows the segmentation-consistency term is the largest single contributor: removing it roughly halves label-swap locality (27.1 to 13.7) and degrades structure (edge IoU 0.41 to 0.30) and per-organ W_1 (0.115 to 0.129). Locality does not fall to a label-free translator's zero — SPADE injects the label at every scale — but the explicit consistency term is what roughly doubles it. Dropping the OASIS LabelMix or the temporal term leaves per-frame fidelity unchanged (their roles are discriminator regularization and frame-to-frame stability).

4.6 Closing the sim-to-real gap for downstream transfer

The near-zero TSTR of Table 1 is a geometry problem, not texture: our substrate renders a *linear* rectangular frame while Kaggle frames are *curvilinear* sector scans, so a segmenter trained on the linear render meets a different spatial prior at test time. Two drop-in fixes (Table 2). **(A)** A curvilinear remap warps the (image, label) render into the measured sector geometry ($\approx 44°$; Appendix Fig. 2) at train and inference, plus probe-motion augmentation: TSTR rises $\approx 4\times$ for SonoSPADE and $\approx 2\times$ for the raw render, tying them (0.099 ± 0.011 vs 0.097 ± 0.006, 5 seeds) — geometry alone moves SonoSPADE from worst to on par. **(B)** Test-time adaptation (TENT) of the downstream segmenter's BatchNorm affine on 50 unlabeled real frames raises SonoSPADE to 0.124 but the raw

Table 2. Downstream transfer (TSTR Dice, higher better) as geometry and test-time adaptation are added. Curvilinear and TENT numbers are the mean ± std over 5 downstream seeds; the linear row is the single deployed run of Table 1. Geometry (A) brings SonoSPADE from worst to tied with the raw render; TENT (B) makes it overtake, on every seed.

Setting	Raw physics ↑	SonoSPADE ↑
Linear render (Table 1)	0.049	0.025
+ curvilinear remap (A)	0.097 ±0.006	0.099 ±0.011
+ curvilinear + TENT (B)	0.106 ±0.006	**0.124** ±0.007

render only to 0.106 (about 3× the gain, because realistic texture makes the synthetic feature statistics adaptable), so SonoSPADE *overtakes* on every one of the 5 seeds. Absolute Dice stays low (kidney and spleen hard, 60-frame test), so this is directional evidence that texture aids transfer once geometry is matched, not clinical-grade segmentation.

4.7 Inference latency: real time, no GPU

The texture stage is built to run inside a reinforcement-learning acquisition loop, so per-frame latency, not peak realism, is the binding constraint. A single generator pass takes 66 ms on CPU (15 frames/s) and 12 ms on an integrated GPU (85 frames/s) at batch one, with *no discrete GPU* (measured, `scripts/bench_latency.py`). A semantic-diffusion synthesizer of higher realism instead runs T denoising passes per frame. Measured on the same machine, a representative ADM/SDM-style denoising U-Net (the family Echo-from-noise [7] builds on; 62M parameters, sized conservatively so it lower-bounds the diffusion cost, and latency is architecture- and step-count-bound, not weight-bound) costs 186 ms/pass on CPU and 26 ms on the integrated GPU — only $\approx 2.7\times$ a SonoSPADE pass, but its T passes make it two orders of magnitude slower *per frame* at a standard T=50 (133× CPU) and up to three at full DDPM (T=250–1000), narrowing to $\approx 66\times$ only under aggressive 25-step DDIM. For a loop of millions of frames, seconds per frame is prohibitive. We cede the whole-frame FID/KID crown for this real-time, GPU-free inference; the realism-first methods [11,7] target offline augmentation where seconds per frame is acceptable, and unlike the equally-fast tissue-agnostic baselines SonoSPADE is per-tissue. We verified this *in situ*: SonoSPADE runs inside a MuJoCo closed-loop ultrasound-acquisition environment, where an imitation-warm-started REINFORCE policy trains and executes against it, rendering a fresh textured frame at every agent step on held-out patients without the texture stage becoming the bottleneck. This is an integration-and-throughput result; the effect on *acquisition quality* is the subject of Sec. 4.8.

4.8 Per-tissue texture improves an in-the-loop acquisition agent

The latency result says the texture stage *can* run in the loop; it does not say per-tissue texture *helps* the agent. Showing that needs a task whose reward depends on per-tissue correctness, scored by a judge that is *fixed across* texture stages rather than matched to each (the flaw that made our earlier goal-reaching benchmark uninformative, Sec. 5). We build one: *organ-conditioned acquisition.* A probe starts with the target organ partially in view and servos to bring it in; the per-step reward is the confidence of a frozen organ recognizer J that the organ is present in the current textured frame. J is trained *only on real* Kaggle frames and masks (never on simulator or SonoSPADE output, so the reward is non-circular) and is the *same* J for every texture stage. The primary metric is texture-independent: the ground-truth fraction of the acquired view that is the target organ, from the true label slice.

First, is the reward viable, i.e. does J recognize the correct organ better under SonoSPADE than under a global recolor? On held-out simulator content J's liver IoU is 0.48 for SonoSPADE versus 0.21 (CUT) and 0.14 (CycleGAN) under a border-cropped J domain-matched to the borderless sim render (0.27 vs 0.07–0.21 under a full-frame J; the ordering is robust to that choice), approaching the real-data ceiling 0.51 (Table 3, 3 seeds; the lead holds on precision too, so it is not a liver-flooding artifact). Notably the untranslated render itself scores 0.27, *above* both recolors: the global recolors actively degrade liver recognition, while SonoSPADE improves on the render — so part of the effect is structure preservation (SonoSPADE also keeps anatomy best, edge IoU in Table 1), not per-tissue texture alone. A fair comparison also requires all three translators trained on the *same* real pool; scoring CUT with an off-pool checkpoint spuriously reverses the result — a reproducibility pitfall we flag and fix. Trained under this reward (imitation-warm-started REINFORCE, 3 seeds, held-out patients), SonoSPADE reaches the highest ground-truth liver visibility, 0.152 ± 0.004, above CUT (0.144), CycleGAN (0.142) and the untranslated render (0.142) from a static start of 0.11, with all three per-seed paired differences positive (+0.003 to +0.015; sign-consistent across 3 seeds, so directional rather than significance-tested). Its reward signal is roughly twice as strong (0.32 vs 0.18–0.20), but the ground-truth margin is *small*: the imitation warm-start and liver's dominance already nearly solve the task. This is a modest, seed-consistent positive — to our knowledge the first evidence that per-tissue texture helps an in-the-loop agent under a fixed external, real-trained judge.

The effect is confined to liver: for kidney, spleen and gallbladder *no* translator (SonoSPADE included) yields a rendering the recognizer identifies (IoU ≈ 0), even rendered prominently with J rebalanced — the same J scores them 0.3–0.8 on *real* data. The bottleneck is the simulator-plus-translator's failure to reproduce their real appearance (kidney lacks cortico-medullary structure; spleen is acoustically near-identical to liver; the anechoic gallbladder is brightened), not the recognizer, so the multi-organ per-tissue claim rests on the offline metrics (locality, organ-W_1) and the in-the-loop gain is shown for the one organ that transfers.

Table 3. Organ-conditioned liver acquisition (3 seeds, held-out patients). Top: a real-trained recognizer J (fixed across texture stages) recognizes SonoSPADE's liver far better than a global recolor's (liver IoU; real-data ceiling 0.51). Middle: trained under J's confidence as reward, SonoSPADE reaches the highest *ground-truth* liver visibility (true label, texture-independent; static start 0.11), higher in 3/3 seeds. Bottom: SonoSPADE's reward signal is $\sim 2\times$ stronger. Harness and the CUT-checkpoint fairness fix in `scripts/premise/`.

	physics	CUT	CycleGAN	SonoSPADE
J liver recognition, IoU ↑	0.27	0.21	0.14	**0.48**
Ground-truth liver acquired ↑	0.142	0.144	0.142	**0.152**
Reward signal (J conf., secondary) ↑	0.20	0.18	0.16	**0.32**

5 Limitations

Unpaired training matches per-tissue statistics, not exact speckle: the output is training texture, not diagnostic imagery. Acoustically similar soft tissues are near-inseparable in B-mode, so per-tissue fidelity is strong only for distinct structures. Validation is on a *single* dataset and anatomy (abdominal Kaggle, 60 annotated real frames, 8 CT cases); external validity to other anatomies and scanners is untested, and the downstream Dice stays low even after the geometry+TENT fixes (kidney and spleen barely transfer), so the transfer result is directional, not clinical-grade. A larger, multi-anatomy matched-view clinical set is the honest next step.

Our evidence that SonoSPADE *improves* an agent is real but bounded. It does not come from the earlier goal-reaching benchmark, whose per-texture *matched* judge is not comparable across texture stages (a random policy scores ≈ 0.36 under one judge versus ≈ 0.03 under another); the organ-conditioned task of Sec. 4.8 fixes that but the gain is small, seed-consistent rather than significance-tested (3 seeds), and liver-only. A harder, less liver-dominated target and a translator that reproduces smaller-organ appearance is the next step to a larger effect.

6 Conclusion

SonoSPADE is a real-time, GPU-free, per-tissue ultrasound texture stage for closed-loop acquisition: conditioning on both a physics render and the tissue label slice makes texture per-tissue and controllable, and a segmenter pretrained free on simulator pairs supplies discriminator supervision and a consistency signal with no paired real labels. A label-swap locality metric quantifies that the conditioning is genuinely per-tissue (≈ 26 versus 0 for label-free translators), addressing a stated evaluation gap. Matching the clinical sector geometry and adapting the downstream segmenter at test time turn a near-zero transfer into SonoSPADE overtaking the untranslated render on every seed, and on an organ-conditioned acquisition task scored by a fixed real-trained recognizer the per-tissue reward gives the highest ground-truth liver visibility in all three seeds — a small, liver-only in-the-loop proof of concept. The whole method trains on a commodity laptop.

When to use it. SonoSPADE is for controllable, label-consistent synthetic data without real annotations: because texture is conditioned on the label, the frame and its mask stay aligned, whereas a global recolor scrambles the per-tissue statistics the labels describe. The recipe and the locality metric transfer to any modality with an aligned-pair simulator. When photorealism alone is the goal a global translator remains the better choice — SonoSPADE is for trustworthy labeled data with per-tissue appearance, not the most realistic single image.

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A Appendix: supplementary tables and figures

Table 4. Whole-frame FID/KID ($\times 10^3$) against the real pool (InceptionV3), lower better. SonoSPADE is not competitive by design (these reward the global recolor it does not perform); the bottom block re-runs in the sector geometry, improving all absolute scores but not the ranking.

Method	FID ↓	KID $\times 10^3$ ↓
Raw physics	315.3	294.3
CycleGAN	**192.2**	203.1
CUT	235.6	245.9
S-CycleGAN (reimpl.)	199.7	**196.9**
SonoSPADE	294.7	306.5
Raw physics-curvi	212.9	265.4
CycleGAN-curvi	**67.9**	**55.6**
CUT-curvi	72.3	55.9
S-CycleGAN-curvi	104.2	84.2
SonoSPADE-curvi	191.1	232.6

Table 5. Ablation (one term off per row; retrained at a fixed budget). Removing the frozen S_{US} seg-consistency term roughly halves locality and degrades structure — the largest single effect.

Variant	Edge IoU ↑	Organ W_1 ↓	Locality ↑
Full SonoSPADE	0.41	0.115	27.1
− S_{US} consistency	0.30	0.129	13.7
− OASIS LabelMix	0.39	0.111	28.6
− temporal consistency	0.42	0.106	26.3

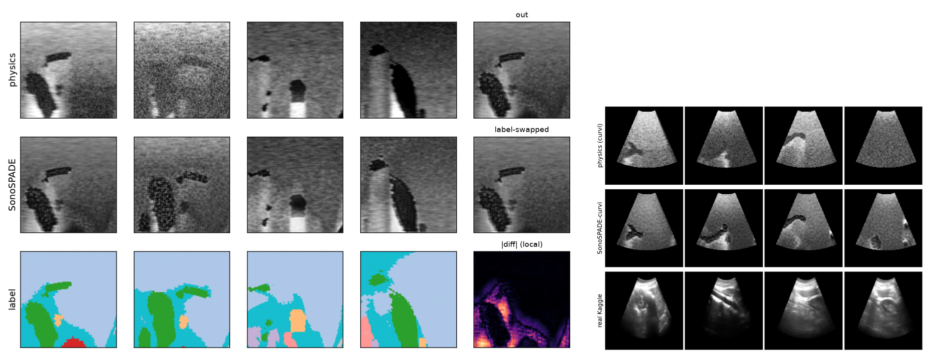

Fig. 2. Left, per-tissue qualitative result (Sec. 4.2): physics render, SonoSPADE output, tissue label (rows); each tissue gets its own speckle, and relabeling a region changes its texture locally. Right, Experiment A (Sec. 4.6): after the curvilinear remap the curvilinear physics render (top), SonoSPADE-curvi (middle), and a real Kaggle frame (bottom) share the sector field.